

Asif Billah

Ph.D. Candidate | Georgia Tech

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🎓 EDUCATION

Present Aug 2024	Georgia Institute of Technology, Ph.D. in ECE ● Focus : Ultra Wide Bandgap Semiconductors ● Supervisor : Dr. P. Douglas Yoder ● Topic : Electronic structure and charge transport within impurity bands in ultra wide bandgap semiconductors ● CGPA : 4.00/4.00
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✂️ WORK EXPERIENCE

Nov 2024	Guest Lecturer, ECE, Georgia Tech.
Nov 2024	● Course Taught : ECE 3040 Microelectronic Circuits
Present	Graduate Research Assistant, Computational Electronics and Photonics Lab
Aug 2024	● Developed theoretical models for shallow donor and deep acceptor impurity bands in ultra wide bandgap materials. ● Working on Monte Carlo charge transport model based on the obtained impurity band dispersions. ● Support for the works are offered by Defense Advanced Research Projects Agency (DARPA).

🔬 RESEARCH EXPERIENCE

Present Aug 2024	Computational Electronics and Photonics Group Dr. P. Douglas Yoder ● Developed theoretical model for shallow donor impurity bands in ultra wide bandgap semiconductors. Manuscripts are in preparation. ● Developing a novel model for deep acceptors in ultra wide bandgap semiconductors. ● Will build a Monte Carlo framework to investigate the carrier dynamics, using the calculated band structures as input.
Jul 2024	Nanophotonics Research Group Dr. Muhammad Anisuzzaman Talukder
Apr 2021	● Explained how element doping, graphene integration, and plasmonic (SPR/LSPR) strategies can overcome absorption, charge-transport, and kinetic limits in transition-metal-oxide photoelectrodes, outlining key performance metrics and future directions for scalable PEC water splitting to produce green hydrogen. [Adv. Energy Mater. (2025)] ● Looked into placing an ultra-thin iridium layer on standard electrodes to improve the efficiency of photoelectrochemical cells' hydrogen evolution reaction using DFT. [I. J. of Hydrogen Energy (2024)] ● Studied enhancing solar hydrogen generation efficiency in PECs by examining the effects of activation overpotential on solar energy conversion to fuel and the choice of suitable photo-absorbing materials and suitable electrode pairs to optimize the process. [Heliyon (2023)]

📖 SELECTED PUBLICATIONS (GOOGLE SCHOLAR)

- 2026 [A. Billah](#), P. D. Yoder, "Impurity band formation in doped AlN : A Green's function approach", **AIP Advances (2026)**. [[Paper](#)]
- 2025 A. A. Mamun, A. H. Chowdhury, [A. Billah](#), J. Karim, A. O. Hussain, F. Rahman, M. A. Talukder, "Advancing Transition Metal Oxide Photoelectrodes for Efficient Solar-Driven Hydrogen Generation : Strategies and Insights", **Adv. Energy Mater. (2025)**. [[Paper](#)]
- 2024 A. A. Mamun, [A. Billah](#), M. A. Talukder, "Enhancing hydrogen evolution reaction using iridium atomic monolayer on conventional electrodes : A first-principles study", **International Journal of Hydrogen Energy (2024)**. [[Paper](#)]
- 2023 A. A. Mamun, [A. Billah](#), M. A. Talukder, "Effects of activation overpotential in photoelectrochemical cells considering electrical and optical configurations", **Heliyon (2023)**. [[Paper](#)]

🧰 TECHNICAL SKILLS

- Programming Languages : C/C++, MATLAB, Python, System Verilog
- Simulation Software : ABINIT, Quantum espresso, VESTA, Lumerical, Ansys, MEEP, COMSOL Multiphysics, Cadence, ModelSim, PSpice, Proteus, AutoCAD, PSAF
- ML Frameworks and Libraries : Deep Neural Network (using TensorFlow and Keras), Image processing (using OpenCV)
- Hardware Skills : Arduino, ATMEGA32, FPGA

RELEVANT COURSEWORK

- Microelectronics Technology
- Advanced Logic Transistor
- Nano-electronics and Nanotechnology, Processing, and Fabrication Technology
- VLSI Circuits and Design
- Engineering Electromagnetics
- Power Electronics
- Optoelectronics
- Gigascale Integration
- Theory of Microelectronics
- Analog Integrated Circuits and Design
- Solid State Devices
- Compound Semiconductor Devices
- Microelectronic Circuits
- Electrical Properties of Materials
- Digital Electronics
- Photonic Devices and Materials

HONORS AND AWARDS

Gold Medal in the [2019 International University Physics Competition](#)

Research grant received from RISE, BUET as an undergrad student

Dean's List Award for academic excellence at undergraduate study

University Merit Scholarship for academic excellence at undergraduate study